CHAPTER 2

Enzymes and Energy

Reflecting Questions

- How do biological processes follow the laws of thermodynamics?
- How do enzymes catalyze reactions and reduce the amount of energy required to fuel them?
- How are enzymes used in industry and pharmaceutical research?

Prerequisite Concepts and Skills

Before you begin this chapter, review the following concepts and skills:

- explaining how a redox reaction works (Chapter 1, section 1.3), and
- describing the structure and function of ATP (Chapter 1, section 1.3).

In the summer of 2001, a forest fire that had been started by a lightning strike raged through Kootenay National Park in British Columbia. Park officials allowed the fire to progress because the area was scheduled for a prescribed burn. Fires are a natural part of forest ecology and are important in forest regeneration. For example, some species of pine, such as the jack pine, drop cones that need the heat from a fire to open them and release their seeds. The Kootenay fire quickly consumed the dry grasses and trees; it soon spread beyond the area park officials could manage, threatening nearby communities.

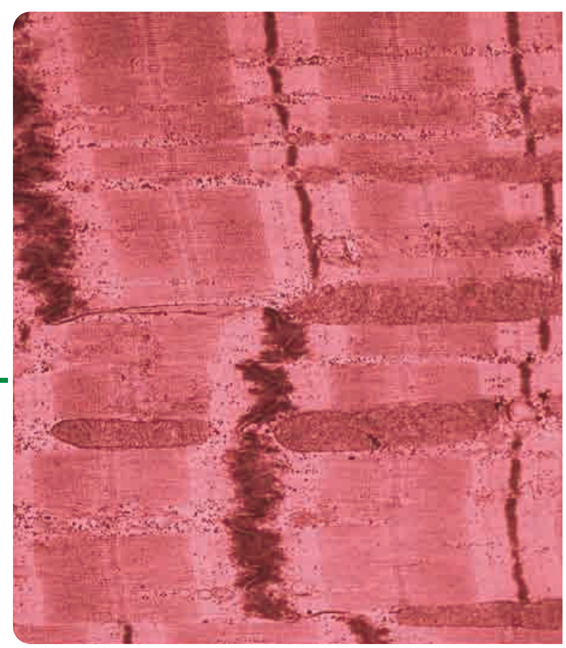
Many firefighters risked their lives to control the spread of the flames. By the time the fire was contained and eventually extinguished, thousands of hectares of forest had burned.

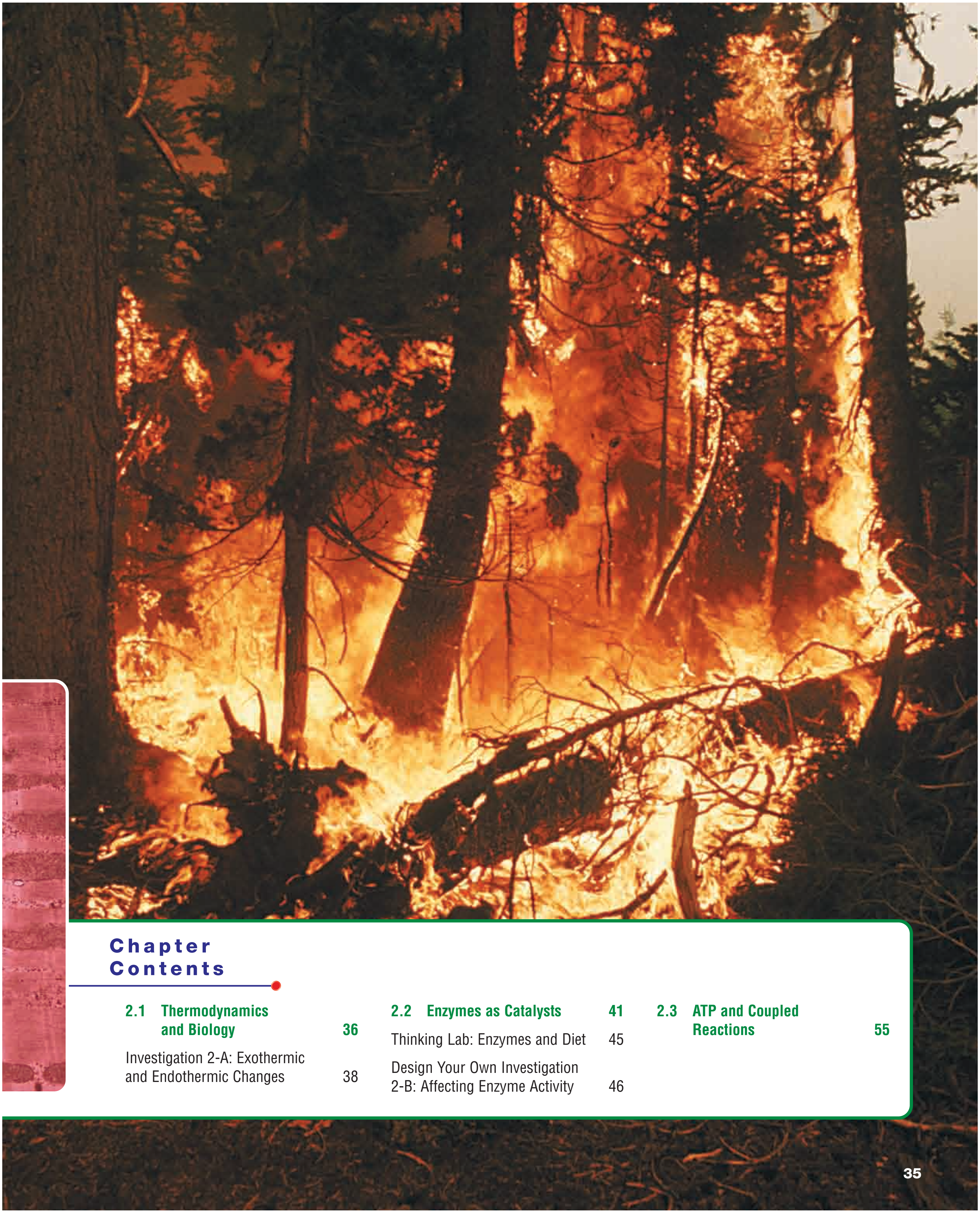
The chemical reaction that occurred in the fire involved oxygen and the wood that formed the trees. While the forest fire was an example of a reaction that occurred with oxygen outside cells, reactions with oxygen also occur inside cells.

Energy is necessary to perform all cellular reactions, including redox, hydrolysis, and condensation reactions. Enzymes aid reactions within cells. Enzymes are necessary because they speed up the synthesis of energy-rich molecules needed for cellular processes.

In this chapter, you will learn how chemical reactions within cells are used to make energy-rich molecules. Energy from these molecules is used for various cellular processes. The bonds that hold atoms together store energy in molecules. This energy can be used by a cell to do work. You will explore various factors that influence how molecular bonds are formed and broken. You will also discover which molecules are involved in cellular processes and how the energy from one reaction can be used to drive another reaction.

How do these heart muscle cells obtain energy to keep the heart beating?





Chapter Contents

2.1 Thermodynamics and Biology

EXPECTATIONS

- Describe the flow of energy in biological systems.
- Apply the laws of thermodynamics to the transfer of energy in the cell.
- Interpret quantitative data to learn how energy can be used in living systems.

Many reactions occur inside every cell. These reactions, collectively known as **metabolism**, have been at the centre of much scientific investigation. For example, manufacturers of dietary supplements for athletes seek to isolate chemicals that increase metabolic activity. Creatine phosphate is one such chemical — it is a nitrogenous molecule that is stored in muscle cells. Enhanced stores of creatine phosphate in muscles have been shown to increase muscle mass and efficiency. The compound was synthesized and used in the former Soviet Union by elite athletes in the 1960s to increase their metabolic activity and performance. What are metabolic reactions, and why are they important? To understand this, you must first understand how energy flows through systems.

Figure 2.1 Metabolic reactions make possible all functions of cells. Some of these reactions fuel the mechanical work that a cell undertakes.

Energy and the Laws of Thermodynamics

To survive, all living things require **energy**, which is the capacity for doing work. Energy comes in different forms. For instance, energy comes from the Sun as light, and thermal energy from a furnace can be used to heat a home. All moving objects, such as falling water and pistons in an internal combustion engine, have **kinetic energy**. Energy can also be stored as **potential energy**. A molecule of glucose has potential energy. The potential energy stored in the bonds of a molecule is called **chemical energy**. If a molecule of glucose is broken down into carbon dioxide and water, the energy released can be used to do work. If a phosphate group is removed from a molecule of ATP, the chemical energy can be used to fuel various cellular processes.

Energy continually flows through living and non-living systems. The study of this flow of energy is called **thermodynamics**. Physicists and chemists have studied thermodynamics since the days of Sir Isaac Newton. Biologists also apply thermodynamics when they study metabolic processes and the energy transformations that take place within living systems. Scientists use the term **system** to identify a process under study, and they refer to it in relation to the rest of the universe. For instance, a hot drink in a sealed vacuum bottle is considered a **closed system** because the liquid is isolated from its surroundings — thermal energy cannot move from the liquid to outside the bottle. Removing the lid from the bottle results in an **open system**, because energy (thermal, in this case) can now move between the liquid and its surroundings — it moves from the liquid to outside the bottle.

All living organisms are open systems; energy moves two ways, both in and out of cells. For example, a green plant absorbs energy from the Sun and uses this energy for building structures, transporting materials, growth, and reproduction. The plant also releases energy into the environment in the form of thermal energy when the plant is forming metabolic products, such as water and carbon dioxide.

How energy flows between organisms and the environment is governed by the **laws of thermodynamics**. You have already encountered these laws in previous studies. The first law, or law of conservation of energy, states that *energy can neither be created nor destroyed, but can be transformed from one form to another*. For example, during photosynthesis, a green plant absorbs light energy from the Sun. This energy is transformed into chemical energy, which is stored in bonds that hold together atoms in a molecule of sugar. An internal combustion engine converts the chemical energy stored in gasoline molecules into kinetic energy — the motion of the car.

Some chemical reactions, such as burning a fuel, release energy. Some of this energy is useful because it is available to do work. The energy available to do work is known as free energy. Free energy can be used to do the work of building molecules in a cell. However, whenever energy is transformed from one form to another, some of it is lost. This lost energy is the portion that is not free energy and therefore is not available for useful work. The amount of free energy that can be harnessed by a green plant or car is much less than the total amount of light or chemical energy present in the sunlight or gasoline. This fact is the basis of the second law of thermodynamics, which states that *energy cannot be transformed from one form to another without a loss of useful energy*. The energy that is lost eventually escapes into the atmosphere largely as waste thermal energy. There are many transformations of energy that occur inside a cell. During each transformation, some energy is lost as thermal energy. Eventually, all forms of useful energy are transformed into thermal energy. After thermal energy dissipates, it can never be transformed back into a useful form, such as chemical energy, that can be used to do work. Therefore, biological systems require a constant supply of energy from the Sun to function.

A measure of the tendency of a system to become unorganized is called **entropy**. Every transformation of energy creates more disorder in the universe. Therefore, we can restate the second law of thermodynamics as follows: *every energy transformation increases the entropy of the universe*.

The conversion of chemical energy into thermal energy does not violate the first law of thermodynamics. If thermal energy is produced during a chemical reaction, it is still a form of energy. Although some of this energy is not available to do work, energy is still conserved.

Consider the following example as a case study of thermodynamic principles. Stacked beside the fire pit at your campsite are a stack of newspapers and a bundle of kindling that you intend to ignite to start a fire. The stack of paper and the wood are composed of cellulose, which is made up of complex carbon-based molecules. These molecules contain potential chemical energy. When you light the paper, the chemical bonds in the molecules are broken in a reaction with oxygen. During the reaction, thermal energy and light are released. Recall from your study of Chapter 1 that this is a redox reaction. Once the reaction begins, the paper quickly burns, forming the products of the oxidation of cellulose: carbon dioxide and water. If energy is released from the reaction of paper with oxygen, the paper and oxygen must contain more chemical energy than the products (see Figure 2.2).

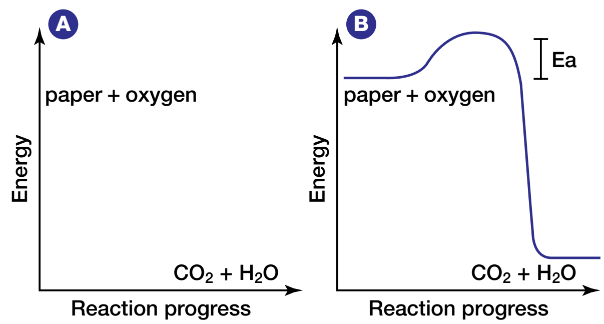


Figure 2.2 (A) Products (carbon dioxide and water) contain less energy than the reactants in a reaction between paper and oxygen. (B) An activation energy, Ea, is needed to initiate the reaction.

During the reaction, the chemical energy stored in the paper and in the oxygen molecules is transformed into thermal energy and light energy. You can feel the thermal energy that is released if you reach toward the fire to warm your hands.

Why does the paper require an initial input of energy to start the fire? Chemical bonds hold atoms and molecules together. These bonds maintain the chemical energy in the molecules. In order to destabilize the bonds, and thereby release the energy they hold, an initial input of extra energy is needed. This extra energy is known as **activation energy**. Figure 2.2 shows the activation energy required to ignite paper. Different substances require different amounts of activation energy to start a reaction. The activation energy needed to start a reaction within cells is governed by special proteins. Without these proteins, metabolic processes could not occur. Next, you will examine two types of metabolic reactions that occur within cells.

Exothermic and Endothermic Metabolic Reactions

Recall that metabolic reactions encompass all the reactions that occur within cells, including anabolic reactions (such as condensation) and catabolic reactions (such as hydrolysis), and redox reactions.

Complex carbohydrates, fats, and proteins can be broken down in catabolic reactions, thereby forming molecules such as simple sugars and amino acids. Anabolic processes then join up these products and their functional groups to form various macromolecules needed by cells for maintenance and growth.

Investigation 2 • A

SKILL FOCUS

- Predicting
- Performing and recording
- Analyzing and interpreting
- Communicating results

Exothermic and Endothermic Changes

The living processes of cells involve numerous chemical reactions involving a wide variety of reactants and products that involve the making and breaking of bonds. Such chemical reactions can be exothermic and release energy or they can be endothermic and absorb energy. Within the cell, there appears to be no net change. In isolation, you can observe how each reaction exchanges energy with the external environment. In this investigation, you will observe energy changes that are characteristic of these reactions.

Pre-lab Questions

- Before a chemical change occurs, where is energy stored?
- How is a cell an open system with respect to energy?
- How is a reaction in an Erlenmeyer flask an open system?

Problem

What energy changes can you observe during different chemical reactions?

CAUTION: The reactants and products formed in this experiment may be toxic. Avoid any contact with skin, eyes, or clothes. Flush spills on your skin immediately with copious amounts of water and inform your teacher. Exercise care when handling very hot or very cold objects. Handle thermometers with care. Wash your hands before leaving the laboratory.

Materials

test tubes
test tube markers
test tube brush
test tube rack
test tube holder
250 mL Erlenmeyer flask
50 mL beaker
250 mL beaker
stirring rods
balance
beaker tongs
alcohol thermometer or temperature probe
squirt bottle
distilled water
ammonium thiocyanate
barium hydroxide (octahydrate)
ammonium chloride
ammonium nitrate
sodium chloride
calcium chloride

Procedure

Part A

1. Measure 9 g of barium hydroxide (octahydrate) and add it to a 250 mL Erlenmeyer flask.
2. Insert an alcohol thermometer into the solid barium hydroxide and record the temperature.
3. Measure 3 g of ammonium chloride and add it to a 50 mL beaker.
4. Using the thermometer or temperature probe, record the temperature of the ammonium chloride.
5. Use a squirt bottle to squirt a few drops of water onto the external surface of the flask. Observe any change in appearance of the water as you continue with the procedure.
6. Add the ammonium chloride to the Erlenmeyer flask containing the barium hydroxide and stir gently to thoroughly mix the ingredients.
7. Immediately insert the thermometer or probe into the mixture and record the temperature. Note any changes in the appearance of the materials in the flask or any other evidence of a chemical reaction occurring in the mixture.
8. Record the temperature every 30 s until there is no change for at least three minutes.

A reaction can be classified based on whether it releases or uses energy. A reaction that is accompanied by a release of energy is called an **exothermic reaction**, as shown in Figure 2.3 on the next page. For example, recall the overall reaction for cellular respiration:

$$\underset{\text{glucose}}{C_6H_{12}O_6} + \underset{\text{oxygen}}{6O_2} \rightarrow \underset{\text{carbon dioxide}}{6CO_2} + \underset{\text{water}}{6H_2O} + \text{energy (useful and waste)}$$

For each molecule of glucose oxidized in cellular respiration, energy is released. Some of this energy is useful and available to do work and

9. Record the temperature and observational data in a table.
10. After all measurements and other observations have been completed, dispose of any chemicals according to your teacher's instructions.
11. Clean flasks, beakers, and thermometers thoroughly with distilled water to remove all traces of chemicals used in this procedure.

Part B

1. Using the balance, measure 3 g of sodium chloride.
2. Pour a small amount of water into a test tube. Record the temperature of the water.
3. Pour the sodium chloride into the test tube and gently swirl the contents until the sodium chloride dissolves in the water.
4. Immediately insert the thermometer into the solution and record the temperature. Note any changes in the appearance of the substances in the test tube or any evidence of a chemical reaction occurring in the mixture.
5. Record the temperature every 30 s until there is no change for at least three minutes.
6. Repeat steps 1 to 5, first using an equal mass of ammonium chloride, and then using an equal mass of calcium chloride. These steps may be performed by other class members or groups.
7. Record your data in a table. Include data of other class members who performed the investigation using the alternate chemical compounds.
8. After all measurements and observations have been completed, dispose of all chemicals according to your teacher's instructions.
9. Clean flasks, beakers, and thermometers thoroughly with water to remove all traces of the chemicals used in this procedure.

Post-lab Questions

1. Describe the energy change of each reaction.
2. Which reaction is endothermic? Which reaction is exothermic?
3. In a cell, energy from one reaction is often used to do work within the cell. Which of these reactions can be used to do work? Justify your choice.
4. The temperature of the surrounding environment may have had an effect on the rate of each reaction. If the temperature in the surrounding environment were very cold, as in a refrigerator, which reaction would be most affected? Justify your response.

Conclude and Apply

5. As mentioned before, cells can maintain a near-constant temperature despite a wide variety of reactions that take place. Some reactions release thermal energy to the surrounding environment, and others absorb thermal energy from the surrounding environment. Design a model cell that incorporates both reactions safely to demonstrate this.

Extend Your Knowledge

6. Examine a "heat pack" and a "cold pack." Identify the chemicals contained in each package. Follow the instructions on the package label to initiate the reaction. Measure the temperature change and time required to attain the maximum or minimum temperature. Observe and record any other change in the properties of the substances in the package. If you know the chemicals involved, write out the endothermic or exothermic chemical reactions that occurred in each package.

some is waste thermal energy. This means that the products (carbon dioxide and water) contain less energy than the reactants (glucose and oxygen).

In contrast, an **endothermic reaction** involves an input of energy. For example, the synthesis of glucose by plants during photosynthesis is as follows:

$$6CO_2 + 6H_2O + \text{energy} \rightarrow C_6H_{12}O_6 + 6O_2$$

carbon dioxide — water — (sunlight) — glucose — oxygen

Synthesis involves building molecules — it uses more energy than it gives off (see Figure 2.3). For example, green plants require energy in the form of sunlight to produce glucose. Because this endothermic reaction stores chemical energy in molecules, there is a gain in energy.

As you can see in the two equations above, oxidation and synthesis of glucose are two reactions that are the reverse of each other. If two reactions are the reverse of each other, one reaction is endothermic and the other is exothermic.

Exothermic and endothermic reactions both involve energy transformations. How do cells control the flow of energy so that they do not overheat and destroy themselves? In the next section, you will learn how cells are able to lower the amount of activation energy necessary to carry out a variety of metabolic reactions.

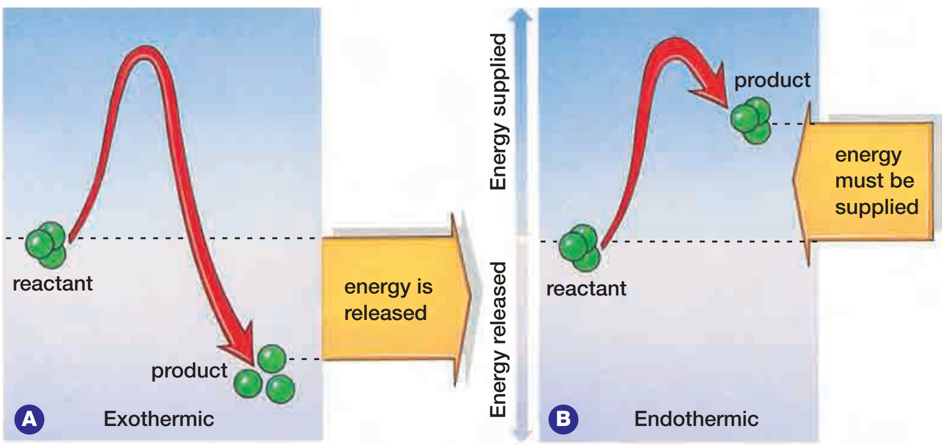


Figure 2.3 In an exothermic reaction (A), the reactants contain more chemical energy than do the products, so there is a release of energy. In an endothermic reaction (B), the reactants contain less energy than do the products, so energy must be added for the reaction to proceed.

SECTION REVIEW

1. K/U How is energy important to a living organism?
2. K/U A molecule of glucose has potential energy.
   (a) What does this mean to a cell?
   (b) Where is the energy in the glucose molecule?
3. K/U Why is a living cell considered to be part of an open system?
4. C Draw diagrams to show how energy flows in an open system and in a closed system.
5. K/U Define the term “free energy.”
6. MC You turn on a CD player and the disc begins to rotate. How is this an example of energy transformation?
7. K/U Define and provide examples of exothermic and endothermic reactions.
8. C An endothermic reaction is often referred to as an “uphill” process. Use an analogy from everyday life to explain why this is a good description.
9. C Make a chart to contrast exothermic and endothermic reactions in terms of energy.
10. K/U Suppose you start a fire by using a magnifying glass to focus the Sun’s image on a piece of paper. What is the source of activation energy for this reaction? What is the source of activation energy after the fire has started and the magnifier is taken away?
11. I A research scientist wishes to study the breakdown of glucose in a solution to form carbon dioxide and water. Design an investigation to determine if this is an exothermic or an endothermic reaction.
12. C Create a flowchart that shows the series of steps that take place starting with reactants and ending with products formed in a chemical reaction. Indicate where energy is involved in the reaction.

UNIT INVESTIGATION PREP

In the investigation at the end of the unit, you will design your own experiment. Be familiar with correct experimental procedure. How will you form your hypothesis? How will you plan your procedure? How will you determine your dependent and independent variables?

2.2 Enzymes as Catalysts

EXPECTATIONS

- **Describe the structure and function of enzymes in cellular respiration.**
- **Design and carry out an experiment involving enzyme activity.**
- **Describe the use of enzymes in the food and pharmaceutical industries.**

Metabolic reactions need activation energy to either build or break down molecules. Cells also use special proteins that aid metabolic reactions. These proteins, called **enzymes**, work by speeding up a chemical reaction. This chemical activity increases the *reaction rate*, or rate at which a reaction occurs, measured in terms of reactant used or product formed per unit time (while existing conditions remain unchanged). Some of the earliest studies on enzymes were performed in 1835 by Swedish chemist Jon Jakob Berzelius, who termed their chemical action "catalytic."

Enzymes and the Catalytic Cycle

The acceleration of a chemical reaction by some substance, which itself undergoes no permanent chemical change, is called **catalysis**. The catalysts of metabolic reactions are enzymes, which are involved in almost all chemical reactions in living organisms. Without enzymes, metabolic reactions would proceed much too slowly to maintain normal cellular functions. Consider the hydrolysis of sucrose, an exothermic reaction. A solution of sucrose dissolved in water could sit for years without showing signs of hydrolysis. If the enzyme sucrase is added to the solution, the enzyme speeds up the reaction millions of times, so that all of the sucrose will be hydrolyzed in several seconds. Enzymes speed up reactions by lowering the amount of activation energy needed. Thus, less energy is required for the reaction to begin. The action of an enzyme on an exothermic reaction is illustrated in Figure 2.4.

Cells carry out a large number of different biochemical reactions; many of these reactions require a specific enzyme in order to take place. Different sets of enzymes are responsible for catalyzing different chemical reactions. **Oxidative enzymes** (oxidoreductases) work to catalyze oxidation-reduction reactions. **Hydrolytic enzymes**

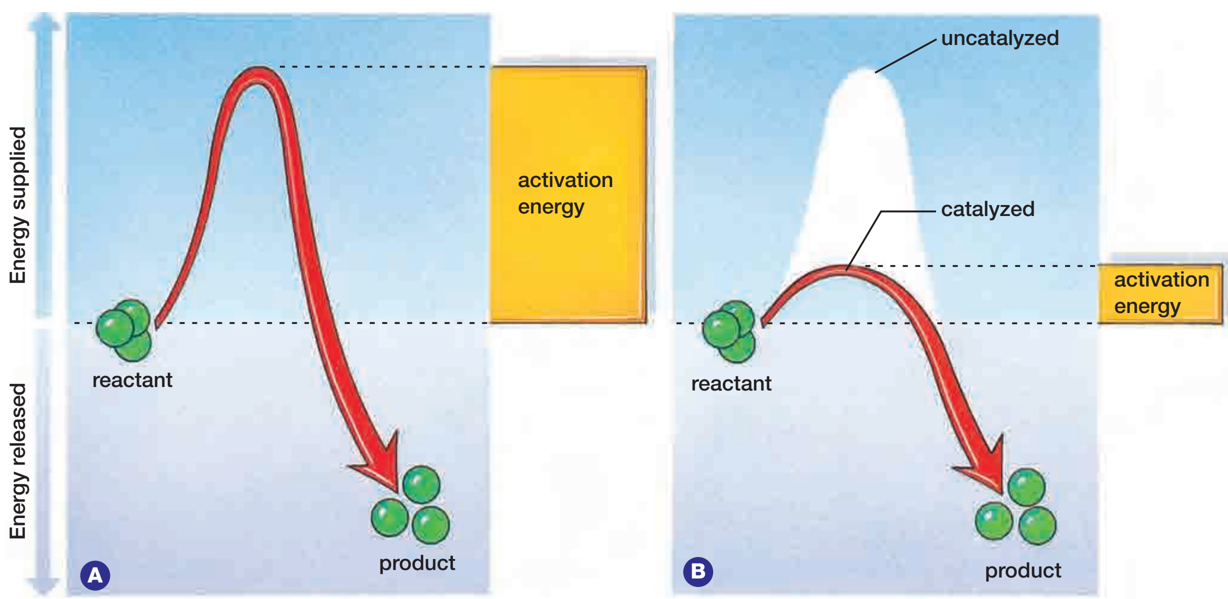


Figure 2.4 (A) Activation energy is required to initiate either an exothermic or an endothermic reaction. (B) Enzymes catalyze certain reactions by lowering the activation energy needed to start the reaction.

(hydrolases) catalyze the addition of water in reactions and split molecules into simpler forms. These simpler molecules may be used to build other molecules or may be excreted from the cell. For example, the lysosomes of cells contain many hydrolytic enzymes. Tasks such as breaking down nucleotides, proteins, lipids, and phospholipids are each carried out by a specific hydrolytic enzyme. Other enzymes remove carbohydrate, sulfate, or phosphate groups from molecules.

Synthesis reactions that build structures such as proteins, nucleic acids, hormones, glycogen, and phospholipids all require the use of enzymes. The enzyme DNA polymerase, for example, is needed for DNA replication, which precedes mitosis. Each chemical reaction in cellular respiration requires a specific enzyme. Deaminases remove the amino groups from amino acids so the remainder of the molecule can be used as an energy source. Enzymes also help to split long-chain fatty acids into smaller compounds, which are used as an energy source and broken down by the process of cellular respiration. Blood clotting, the formation of angiotensin II to increase blood pressure, and the transport of carbon dioxide in the blood all require specific enzymes. Tables 2.1 and 2.2 show categories of enzyme specificity and modes of action.

Table 2.1
Categories of enzyme specificity

Enzyme specificity	Action
absolute	catalyzes only one reaction
group	acts only on molecules that have specific functional groups
linkage	acts only on a particular chemical bond, regardless of the rest of the molecular structure
stereochemical	acts only on a particular optical isomer

A reactant in any given enzymatic reaction is called a **substrate** for that specific enzyme. Some enzymes catalyze one individual reaction; this is the case with peroxidase, an enzyme that decomposes hydrogen peroxide into water and oxygen. Reactions within cells, however, are often part of a **metabolic pathway** (series of linked reactions), beginning with one substrate and ending with a product. Such metabolic pathways can involve many reactions, which often include other pathways. Each step of a metabolic pathway, or each constituent reaction of the pathway, needs its own specific enzyme.

Table 2.2
Enzymes are classified (by an International Classification of Enzymes) according to the kind of chemical reactions they catalyze. The six classes of enzymes and their actions are given.

Action	Type of enzyme	Examples
add or remove water (hydrolysis or condensation)	hydrolases and hydrases	hydrolases: esterases, carbohydrases, nucleases, deaminases, amidases, proteases hydrases: fumarase, enolase, aconitase, carbonic anhydrase
transfer electrons (redox reactions)	oxidoreductases	oxidases, dehydrogenases
split or form a C–C bond	transferases	desmolases
change geometry or structure of a molecule	isomerases	glucose phosphate
form carbon to carbon, carbon to sulfur, carbon to oxygen, or carbon to nitrogen bonds by condensation and hydrolysis reactions coupled to ATP	ligases	pyruvate carboxylase, DNA ligases
add groups to a C=C double bond or remove groups to form a C=C double bond	lyases	aldolase, decarboxylases

To understand how enzymes work, consider that the key to enzyme function is enzyme structure. Enzymes are globular proteins with depressions on their surfaces, as shown in Figure 2.5. These depressions are called **active sites**. Active sites are places where substrates fit and where catalysis occurs. Active sites are not static receptacles. Substrates fit closely into active sites because enzymes can adjust their shapes slightly to accommodate the substrate. This process involves a subtle change in conformation, or three-dimensional shape, of the enzyme when the substrate binds to it. Multiple weak bonds between the enzyme and the substrate are involved in this process. The change in shape of the active site to accommodate the substrate is called **induced fit**. This process may bring specific amino acid functional groups on the enzyme into the proper orientation with the substrate to catalyze the reaction (see Figure 2.5 on the next page).

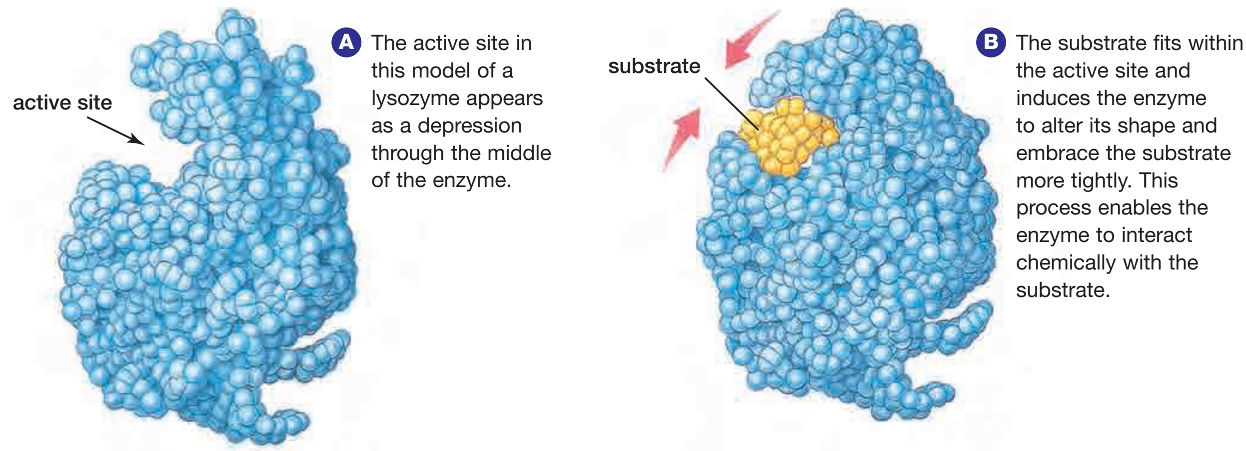


Figure 2.5 The induced-fit model of enzyme action

The combination of the substrate and the enzyme itself forms a compound called an **enzyme–substrate complex**. Swedish chemist Svanté Arrhenius first hypothesized about the enzyme–substrate complex in 1888, proposing that there must be a stage during catalysis when the enzyme and the substrate join together. Modern laboratory experiments have confirmed his hypothesis. In many cases, the enzyme–substrate complex is held together by such bonds as hydrogen bonds and weak ionic bonds. The polar and non-polar groups of the active site attract compatible groups on the substrate molecule. These attractions effectively lock the substrate molecule in the active site. Once in the active site, the substrate is subject to necessary collisions, bond breaks, and bond formations that must take place to form the product molecule. This reaction can be anabolic or catabolic, depending upon the enzyme. Once the product molecule has been formed, it is released from the enzyme–substrate complex. The enzyme is now able to accept another substrate, and begin the process anew. This cycle is known as the **catalytic cycle**. Figure 2.6 shows the catalytic cycle involving sucrose and the enzyme sucrase.

There are several methods by which enzymes reduce the activation energy needed to break the bonds in a substrate. In the enzyme–substrate complex, the substrate molecules experience

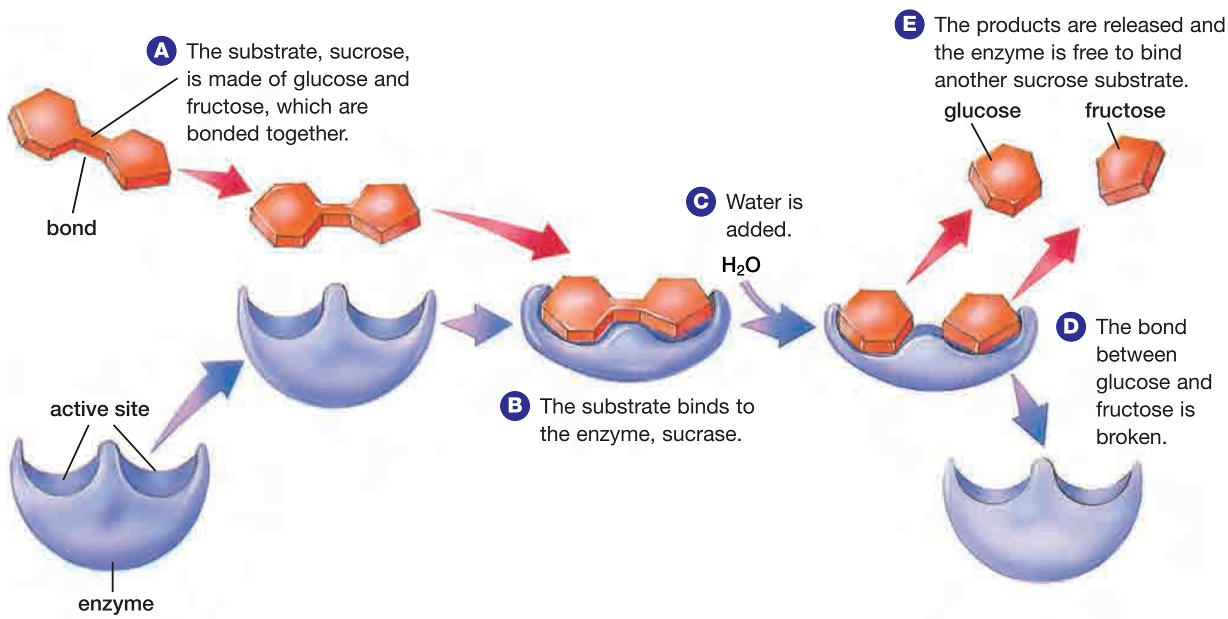


Figure 2.6 The catalytic cycle of this enzyme splits the sugar sucrose into two simpler sugars, glucose and fructose.

physical stress. The *R*-groups in the active site of an enzyme are able to stress the bonds of the substrate. There is bending and stretching of bonds that hold the molecule and the active site together. In this case, the activation energy is lowered because the bonds within the molecule have become weaker, reducing the amount of energy needed to break them.

Another way in which the active site of an enzyme may lower activation energy involves special amino acids that line the active site. These amino acids have reactive *R*-groups that can aid in the transfer of hydrogen ions to or from the substrate. For example, the active sites of hydrolytic enzymes, such as those within the lysosome, often provide acidic and/or basic amino acid groups at precisely the correct orientations required for catalysis. The yeast enzyme, intertase (also known as beta-fructofuranosidase), is a hydrolytic enzyme that speeds up the breakdown of sucrose into the products glucose and fructose.

Some other enzymes provide amino acid groups at their active sites that can accept electrons, while others are attracted to atomic nuclei of the substrate. This process can form a temporary attraction with the substrate. In this state, the substrate is less stable and can more easily react to form the product. Some enzymes may facilitate the correct reaction by bringing two different substrates together in the appropriate orientation to each other.

An oxidative enzyme (such as cytochrome P450s) catalyzes the transfer of electrons from substrates to oxygen molecules. Substrates for these enzymes are often referred to as hydrogen donors because hydrogen ions along with electrons are taken from the substrate. Cytochrome P450s is most common in the endoplasmic reticulum of liver cells. In these cells, the enzyme helps to metabolize toxins, as well as fat-soluble vitamins such as A, D, and E.

WEB LINK

www.mcgrawhill.ca/links/biology12

To find out more about lysosomal enzymes, go to the web site above, and click on **Web Links**. Prepare a three-column chart with the following headings: enzyme, substrate, product of reaction. Fill in your chart as you read about enzymes and their function. Predict what would happen if any of the lysosomal enzymes did not function properly.

Enzyme Activity

As you have learned, enzymes lower the activation energy required to start a chemical reaction. The activity of enzymes, however, can be influenced by environmental factors, such as pH and temperature.

The shape of an enzyme is determined by hydrogen bonds, which hold peptide chains in the enzyme in a specific orientation. (See Appendix 5, “Shapes of Selected Macromolecules,” on page 559.) As well, all enzymes contain segments that are hydrophobic. The hydrogen bonds in an enzyme and any hydrophobic interactions that parts of the enzyme may experience are easily affected by changes in temperature. Enzyme activity increases as temperature increases, but only up to a maximum point. If the temperature increases beyond a critical point, enzyme activity declines rapidly (see Figure 2.7). When this occurs the enzyme has been denatured. When an enzyme is denatured by excessive heat, its shape changes and it can no longer bind to its substrate.

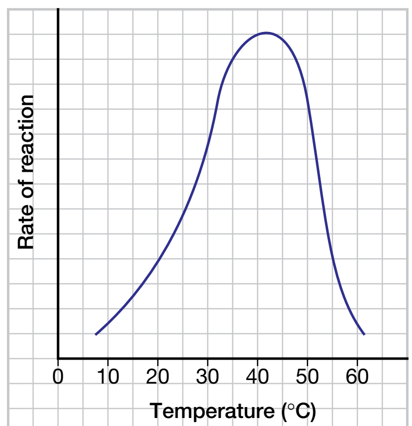


Figure 2.7 Rate of an enzymatic reaction as a function of temperature. The rate of the reaction shown here approximately doubles with every 10°C increase in temperature. For this reaction, the maximum rate occurs at 42°C; the rate then decreases and the reaction eventually stops.

Most human enzymes function best between 35°C and 40°C. Below this temperature range, enzymes are less flexible and therefore less able to provide an induced fit to substrates. Above this range, the bonds become weaker and less able to hold the peptide chains in the enzyme in the proper orientation. Some bacteria, however, can function at temperatures as high as 70°C. These bacteria live in and around **hydrothermal vents**, which are fissures in the Earth’s crust on the ocean floor that release hot water and gases. The bacteria are able to survive in these environments because